A4: IIoT Architecture Poster

Context and Rationale

Manufacturing Setting: The Ubuntu Naturals Antananarivo baobab primary processing plant (Madagascar, Tier 3; Stage 2) is the reference site for a Tier-3 plant-family IIoT architecture spanning Lilongwe (moringa, Malawi), Antananarivo (baobab, Madagascar), Dar es Salaam (hibiscus, Tanzania), and Accra (African ginger, Ghana). A regional fibre backbone connects the four of them, and that is roughly where the shared resource ends: each site keeps only a handful of on-site IT staff, loses power 2-3 times in a normal week, and still has to satisfy EU export traceability, because 74.6% of Ubuntu's raw-material input is organic-certified.

Rationale: Everything downstream -- encapsulation, tableting, liquid-fill -- inherits whatever raw-material quality Ubuntu's Tier-3 plant family sets upstream. Those plants have no monitoring infrastructure worth the name, so they cannot answer an EU due-diligence question and cannot hold primary-processing quality steady across four different botanicals. Left to solve it one site at a time, each plant buys what it can afford: fragmented solutions that will not interoperate, or a clipboard and a manual inspection that no EU customer audit will accept. Drawing the IIoT architecture once, for the constraints the Tier-3 family actually has -- frequent power outages, limited IT expertise, shared regional infrastructure -- gives Ubuntu plant engineers and regional management something to work from when they make procurement, implementation, and integration decisions across all four Tier-3 sites.

Brief

Design an end-to-end IIoT architecture for the Antananarivo baobab primary processing plant, and put it on a poster good enough to hang at an industry conference or at an internal Ubuntu Naturals technology review. Three sorts of people will stand in front of it -- plant engineers, regional IT staff, and senior co-operative management -- and the poster has to work for all three. Design it so that the three sister Tier-3 plants could build from the same sheet.

Key Requirements

- Design a layered IIoT architecture diagram showing: sensor/device layer, edge/gateway layer, network/communication layer, cloud/platform layer, and application/analytics layer
- Specify appropriate sensors for primary processing (drying temperature/humidity, grinder load/vibration, sieve throughput, ambient conditions for organic certification)

- Justify the protocol you pick at each layer -- Modbus RTU at field level, MQTT for edge-to-cloud, REST API at the application layer, or your own alternatives -- against the power reliability and the bandwidth you actually have
- Include an edge computing strategy that addresses the 2-3 weekly power outages (local buffering, store-and-forward, graceful degradation)
- Map at least 4 Industry 4.0 pillar technologies to specific functions in the architecture (e.g., Big Data for quality analytics, Cloud for centralised dashboards, IIoT for sensor networks, System Integration for ERP connectivity)
- Add a data flow diagram tracing how a raw sensor reading becomes something a person acts on, tied back to the data-information-knowledge-wisdom pyramid
- The poster has to read on its own: clear, labelled, and comprehensible with nobody standing beside it to explain

Production Guidelines

1. **Define the scope and layers**: Fix the five layers of the IIoT architecture first (sensor/device, edge/gateway, network/communication, cloud/platform, application/analytics), then sketch the poster layout and check that every layer has room to be read.
2. **Specify sensors per production step**: Go through baobab primary processing a step at a time and find the sensors each step needs -- drying (temperature, humidity, airflow, moisture content), grinding (motor load, vibration, particle-size sensing), sieving (throughput, mesh-vibration), and ambient/storage (humidity, temperature, which is what organic-certification compliance turns on). List manufacturer types or generic specifications.
3. **Select and justify protocols at each layer**: Choose a protocol for each boundary -- Modbus RTU or IO-Link at field level, MQTT or OPC UA at edge-to-cloud, REST APIs or GraphQL at the application layer. Annotate each choice on the poster in a line or two, and tie the reason back to the Tier-3 family's constraints.
4. **Design the edge computing strategy**: Say plainly what happens during the 2-3 weekly power outages. Show local data buffering, store-and-forward mechanisms, graceful degradation modes, and how the system recovers afterwards. UPS-backed edge gateways are worth costing.
5. **Map Industry 4.0 pillar technologies**: Pick at least 4 of the 9 Industry 4.0 pillars and show where each one lives in the architecture -- Big Data analytics in the cloud layer, a simulation or digital twin for drying-tunnel recipe optimisation, system integration reaching the Nairobi ERP hub.
6. **Create the data flow diagram**: Follow one signal the whole way: raw sensor reading, edge processing, cloud storage, analytics, and finally a dashboard someone acts on. Map that route onto the DIKW pyramid (data, information, knowledge, wisdom).
7. **Design for visual clarity**: Use a layout tool you can control (Canva, PowerPoint, Illustrator, draw.io). The poster has to be readable with nobody standing next to it: clear headings, a flow that runs one way (top-down or left-right), consistent iconography, font sizes still legible at A1 scale, and a colour scheme that tells the layers apart.
8. **Export as A1 landscape PDF**: Check resolution and readability at actual print size rather than zoomed to fit; include a title block with the plant name, date, and author.

Marking Scheme

Criterion	Weight	Fail (0-39)	Pass (40-59)	Merit (60-69)	Distinction (70-84)	Stretch (85-100)
Architecture Design	25%	Architecture is incomplete or structurally flawed; layers missing or confused; no clear separation of concerns.	Five layers identifiable; basic components shown at each layer; architecture is logically sound but generic (not adapted to the Antananarivo / Tier-3 scenario).	All five layers well-defined with components specific to the Antananarivo plant (correct sensor types, primary-processing-relevant analytics); clear boundaries between layers.	Architecture addresses Tier-3-family-specific constraints throughout (shared regional fibre backbone, limited per-plant IT staff implications for managed services, EU due-diligence requirements shaping data retention); demonstrates systems thinking.	Architecture includes a scalability dimension (how to extend to the other three Tier-3 plants -- Lilongwe, Dar es Salaam, Accra); addresses multi-tenant considerations for shared infrastructure; proposes a reference architecture the Ubuntu Naturals regional ops team could standardise across the family.
Sensor and Protocol Specification	20%	Sensors vague or incorrect for primary-processing operations; protocols not specified or inappropriately chosen.	Sensors plausible for each production step; at least 2 protocols specified; basic justification for choices.	Correct sensor types for drying, grinding, sieving, and ambient/storage with reasonable specifications; 3+ protocols specified with justification referencing bandwidth, reliability, and power constraints.	Protocol choices include detailed consideration of determinism requirements (drying-tunnel temperature control), data volume (high-frequency grinder vibration monitoring), and standards compliance (OPC UA for interoperability).	Includes a protocol comparison matrix; specifies data rates and payload sizes; addresses protocol security (TLS for MQTT, certificate management for OPC UA) for EU traceability audit requirements.
Edge Computing and Resilience	20%	No edge strategy; power outage problem not addressed.	Edge layer shown; brief mention of local buffering for power outages but no detailed design.	Store-and-forward mechanism clearly described; UPS or battery backup shown for edge gateways; graceful degradation modes identified (what continues locally during an outage).	Edge strategy includes recovery sequencing (what syncs first when power returns), data integrity checks for buffered data, and bandwidth management for catch-up synchronisation; addresses data loss prevention quantitatively.	Proposes a tiered edge architecture (lightweight edge at each major process step, aggregation gateway per line, plant edge server) with defined processing at each tier; includes cost and power consumption estimates; addresses edge firmware update strategy for IT-lean environments.

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Industry 4.0 Pillar Mapping	15%	Fewer than 2 pillars mapped, or mapping is incorrect/superficial.	At least 3 pillars mapped to functions in the architecture; mapping is correct but not deeply integrated.	4+ pillars mapped with specific examples for the primary-processing context (e.g., "simulation for drying-tunnel recipe optimisation to reduce energy use and protect baobab nutrient retention").	Pillars are woven into the architecture rather than listed separately; connections between pillars shown (e.g., Big Data feeds simulation models, which drive autonomous optimisation in the cloud layer).	All applicable pillars mapped with a maturity model showing current state, target state, and implementation sequence; includes pillars relevant to the Madagascar / Tier-3 context (e.g., additive manufacturing for spare parts in import-constrained environments).
Visual Design and Communication	20%	Poster is illegible, cluttered, or visually incoherent; would not communicate the architecture without verbal explanation.	Poster layout is basic but readable; logical flow exists; labels present but formatting inconsistent.	Poster is visually clear with consistent colour coding for layers, legible at A1 scale, logical flow, and professional appearance; comprehensible without a presenter.	Poster uses sophisticated visual hierarchy: icons, colour zones, data flow animations (implied via arrows and sequence numbering); includes a legend; would be appropriate for an industry conference.	Poster achieves infographic quality: balances technical detail with visual impact; includes a summary callout box for executive audiences; uses whitespace effectively; consistent iconography throughout; accessibility-conscious colour palette (colour-blind safe).

Stretch Goal

An exceptional poster would include a companion implementation timeline showing a phased rollout (pilot on one Antananarivo drying tunnel, then the rest of the Antananarivo line, then replication across Lilongwe / Dar es Salaam / Accra), with estimated costs and ROI projections. It would address the Tier-3-family dimension -- how the architecture becomes a shared service for the four sister plants -- and include a governance model for shared data infrastructure. The poster would also show how the architecture enables EU traceability requirements (farmer-lot-to-finished-pack audit trail) and propose a digital twin concept for drying-tunnel optimisation that could reduce energy consumption.

Submitting This Artefact

What you hand in: A1-size digital poster (PDF format, landscape orientation) suitable for conference-style presentation.

This is a Phase 2 (Applied) artefact. It builds on A1 (automation fundamentals), and the work you do here carries into A9 (IIoT within full process map) -- so it is worth doing properly the first time rather than twice.

This brief is attached to the assignment as `A04_iiot_architecture_poster.pdf`; you work from it plus the lecture and workshop materials it names. A worked exemplar from another Ubuntu Naturals site is attached too, with the marker's note on it, and `A04_genai_guide.pdf` shows how to build your own submission with a free Claude account.

Name the file `A04_IIoT_Architecture_Poster_[YourSurname].pdf`, with your own surname in place of `[YourSurname]` -- the portfolio guide uses this convention throughout, and a marker looking for your work will look for that name.

Whatever else it contains, the file needs type that is legible at the specified size; colour-accessible (avoid red-green as sole differentiator).